

(a) Intersector rate $\eta\Gamma_0$; intrasector rate scale Γ_0



Both pairs conduct; only A has direct Hall readout at zero passive mass.

(b) Fixed order of added physical degrees of freedom

1 Active reference current

$$g_{A0}/\|g_{A0}\|$$

Full uncoupled current response

2 Active valley distortion

$$(I - u_1 u_1^T) p_A$$

Orthogonal valley change

3 Passive current

$$b_P/\|b_P\|$$

Driven longitudinal motion

4 Passive valley population

$$p_P/\|p_P\|$$

Opposite partner populations